Technical Appendix

Making Mental Models Scalable with AI:
The `MindGraph` Method

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This appendix documents the instrument described in the Stage-1 proposal in enough detail to evaluate and reproduce it. Section 2 describes the pipeline; Sections 3–4 the taxonomy and the prompt; Section 5 the validation design and metrics; Section 6 the benchmark results; Section 7 the full-sample deployment; Section 8 the accompanying repository. Survey responses, human codings and model outputs are not publicly released; the repository contains synthetic examples on which every script runs.

1 Existing methods for measuring mental models

Table 1 compares the instrument with the methods used so far to measure mental models in organizational research and economics.

2 Pipeline architecture

2.1 Overview

The instrument codes open-ended survey responses with a command-line program calling the OpenAI API. Input: the respondent's answer to *"What do you think causes climate change?"* Output: a directed acyclic graph (DAG) whose nodes are factors from a 25-factor taxonomy (Section 3) and whose unique sink is the outcome node *Climate Change* (ID 999).

2.2 Model call

Each response is coded by one call to a reasoning model. The live configuration is `gpt-6.1-terra` at its default reasoning effort. The system prompt is a fixed text file into which the taxonomy is injected verbatim at a placeholder; the user message contains only the respondent's text. Output is requested as strict JSON with no surrounding prose. The per-call output budget is 4,000 tokens, raised to 10,000 on one automatic retry if the first response is truncated or fails to parse. Calls are stateless: no information from any other response enters the context, so the instrument cannot learn a distribution from earlier items and impose it on later ones. Responses may be in any language; the prompt instructs the model to code on meaning regardless of language.

2.3 Versioning and de-duplication

Every annotation is stored in a cumulative registry keyed by respondent under a `coding_version`: a 12-character SHA-256 prefix of the full system prompt (taxonomy included) concatenated with

Table 1: Methods for measuring mental models

Method (examples)	Content supplied by	Structure captured	Cost and scale	Main limitation
Likert questionnaires, scenario ratings (Levesque et al. 2001; Webber et al. 2000)	Researcher (items)	None: content and agreement only	Low; scalable	Not a structural measure; unrelated to team process ($\rho \approx -.05$)
Paired-comparison relatedness ratings + Pathfinder / MDS / UCINET (Mathieu et al. 2000; Lim & Klein 2006)	Researcher (fixed concept list)	Undirected network; meaning of "related" (causal, co-occurrence) unspecified	$n(n-1)/2$ judgements (14 concepts = 91); fatigue; mostly lab	Concepts imposed; no causal direction
Concept mapping (Marks et al. 2000; Ellis 2006)	Researcher (concepts, often the structure)	Pre-specified, often sequential or hierarchical	Manual scoring; laborious for large maps	Structure may not exist in respondents' minds
Card sorting (Smith-Jentsch et al. 2001, 2008)	Researcher (concepts); respondent groups them	Clusters; no direction	Tedious with many concepts; cumbersome scoring	Concepts imposed; groupings are not causal links
Repertory grids (Reger & Huff 1993; Hodgkinson 1997; Wright 2004)	Respondent (personal constructs)	Dimensional or hierarchical; not causal	Interview-based	Hard to compare across respondents
Causal / cognitive mapping from interviews or documents (Axelrod 1976; Eden 1992; Markóczy & Goldberg 1995; Hodgkinson et al. 1999)	Respondent, or a pool of researcher constructs	Directed cause–effect graph	Manual elicitation and coding; small samples	Coder judgement hard to reproduce; single-language
Hand-coded text: qualitative coding and DAGs from open-ended surveys (Carley 1997; Kaplan 2008; Andre et al. 2024)	Respondent's own words	Concepts and links coded by researchers; directed if coded so	Trained coders; double coding on subsamples (e.g., 200 responses)	Heavy coder influence; slow; agreement 88% factors, 77% links (Andre et al.)
Computational text analysis: topic models, co-occurrence, embeddings, agent–verb–patient extraction (Hannigan et al. 2019; Ash et al. 2021; Lange et al. 2022)	Corpus text	Associations or event triples; causal direction rarely identified	Fully automated; scalable	Association is not causation; not benchmarked against human coding of structure
This proposal: taxonomy-constrained LLM extraction	**Respondent's own words**, any language; nodes mapped to an *a priori* taxonomy	**Directed causal DAG**; only links the respondent states	**About $0.005 per response**; 2,964 responses in 12 countries coded	Needs a domain taxonomy; explicit content only; human benchmark so far EN and FR

Notes: Correlations from DeChurch & Mesmer-Magnus (2010); technique characteristics from Mohammed, Ferzandi & Hamilton (2010, Table 2), Lim & Klein (2006) and Becker et al. (2025). For the proposed method, three-way agreement (LLM and two trained coders, $N = 30$ per language) is Fleiss' $\kappa = .64$–$.65$ on edges and .73–.77 on nodes in English and French, with LLM–human agreement at or above human–human agreement; each coding run is tagged with a prompt-and-taxonomy version hash.

the model identifier. A run skips any respondent already present under the current hash with a non-null annotation; a change to the prompt, the taxonomy, or the model therefore invalidates prior results rather than silently reusing them. A `-force` flag re-codes everything; a respondent whose stored annotation is null (a failed call) is always retried. Prompt files are under version control, so any `coding_version` can be traced to an exact prompt text.

2.4 Batch mode and cost

Runs can be submitted through the OpenAI Batch API (asynchronous, 50% price reduction). On submission the exact list of items is written to a manifest; resumption reloads from the manifest rather than recomputing from the input, so results cannot be misaligned if the registry changes while a batch is in flight. Based on measured token usage in the benchmark runs, batch coding costs about $0.003 per response.

2.5 Tooling

- `narrative_gpt.py`: command-line pipeline. Flags: `-input`, `-output`, `-batch`, `-resume`, `-force`, and `-export-registry`.
- `streamlit_app.py`: web interface.
- `annotation_app/app.R`: R Shiny application in which human annotators code the same items under the same taxonomy; DAGs are built edge by edge with a factor glossary. Output format is identical to the model's, so human and model codings are directly comparable.
- `Ancillary/benchmark_agreement.py`: computes all agreement statistics in Section 5.

3 Causal factor taxonomy

The taxonomy was written before data collection from pilot responses, the climate-science literature, technical reports, and press coverage, and refined in pilot coding sessions. Each factor has a definition and example statements (full text in the repository). There is no factor #23.

ID	Factor	Scope note
Natural causes		
1	Climate cycles	Glacial/interglacial rhythms
2	Solar activity & orbital changes	Solar radiation, Milankovitch cycles
3	Volcanic activity	
4	Ocean cycles	El Niño/La Niña, AMOC
5	Geological & tectonic movements	
Human-caused (I): structural and systemic		
6	Greenhouse gases	CO_2, methane, heat-trapping gases; excludes generic "pollution"
7	Overproduction & output pressure	Profit/output maximisation at sector or firm level; supply chains; data centres

ID	Factor	Scope note
8	Energy production & fossil-fuel dependency	Supply side only: extraction, power plants, refineries, grids
9	Industries & industrial processes	Factories, cement, steel, fertilisers, industrial waste
10	Overexploitation of ecosystems & land use	Deforestation, mining, urbanisation, loss of sinks
11	Agriculture & livestock	Methane, fertilisers, pesticides, land conversion
12	Military emissions & warfare	
Human-caused (II): personal and cumulative		
13	Consumerism & overconsumption	Demand for goods, fast fashion, packaging
14	Dietary choices	Meat, dairy, imported food
15	Household energy choices	Heating, insulation, appliances
16	Transportation choices	Private cars, flying, non-EV purchase
17	Waste production & management	Individual waste, recycling, food waste
Political and institutional responsibility		
18	Governmental inaction	Weak or unenforced regulation, failed agreements
19	Lack of low-impact transport systems	Public-investment failure (contrast #16)
20	Corporate influence & lobbying	Corporations as emitters and political actors; greenwashing
21	Economic growth	Capitalism, GDP logic, industrialisation as a system, globalisation; includes population growth
22	Inequality & environmental injustice	
Skepticism, misinformation, alternative theories		
24	Scientific weaponisation	Engineered climate, chemtrails, HAARP
25	Government control	Climate change as pretext for control
26	Data manipulation & media complicity	
999	*Climate change*	Outcome node; may only be a target

4 Cause-narrative prompt

4.1 Task structure

The system prompt casts the model as an expert in climate change and causal reasoning performing a classification task, and specifies three ordered tasks:

1. **Node identification.** Identify which taxonomy factors the text mentions, using taxonomy

IDs and labels only. If none is identifiable, return an empty graph `{"nodes": [], "edges": []}`.

2. **Edge assessment.** Define directed links among the identified nodes as if drawing a DAG. Climate Change (999) is always the final node and can never be a source. No nodes may be added at this step. Links must be coherent with the text; links not stated in the text are not to be inferred.

3. **Coherence check.** Verify that 999 is present and terminal, and correct any incoherence in place before output.

Output is a JSON object with a `nodes` array (`id`, `label`) and an `edges` array (`source`, `target`).

4.2 Annotation guidelines

Eleven rules resolve recurrent ambiguities. They were written from pilot and benchmark disagreements.

(a) Pure lists. Factors listed without stated links each cause 999 directly.

(b) Greenhouse gases (#6). Use #6 for explicit mention of greenhouse gases, CO_2, methane, heat-trapping gases, or atmospheric pollution; not for generic "pollution." Insert #6 as an intermediate node *only* when the text names the gas mechanism ("releases CO_2"); otherwise code the direct edge. This is the parsimony rule.

(c) Energy (#8) vs. sectors. #8 only for supply-side energy systems. Fossil fuels used by a named sector or individual go to that sector (#16 vehicles, #9 factories, #15 households).

(d) Individual vs. systemic. Individual waste #17 vs. industrial waste #9; overproduction #7 vs. economic growth #21; individual transport #16 vs. missing transport infrastructure #19.

(e) Only the respondent's own causes. Do not code causes the respondent attributes to other people's beliefs, or historical events not presented as current causes.

(f) Parenthetical examples count. Factors cited as examples inside parentheses are coded.

(g) Pure descriptions. Effects-only or belief-only text returns an empty graph.

(h) Inaction caused by skepticism. Code the chain skepticism factor $\rightarrow$ 18 $\rightarrow$ 999, not a single node.

(i) "Naturally" as "obviously" does not trigger natural causes.

(j) "Industrialisation" refers to the system: #21, not #9 or #7.

(k) "Profit maximisation": #21 if the economic system is meant, #7 if a sector or firm.

5 Validation design and metrics

5.1 Benchmark sample and coders

Two independent benchmark samples of $N = 30$ responses each: English responses from an online panel (Prolific, United States/United Kingdom) and French responses from a laboratory session. Two trained human coders (Coder 1 and Coder 2; both economists working on climate policy) coded every item in the Shiny application, independently, without communication and without access to model output. The model coded the same items.

5.2 Units of analysis

Node level. For each respondent, a binary vector of length 25 (one entry per factor). The universe is fixed, so the measure does not depend on graph structure.

Edge level. For each respondent, the universe is the set of all directed pairs among the nodes mentioned by *any* rater (plus 999 as target); each potential edge is scored present/absent per rater. This is the primary measure of agreement on causal structure.

5.3 Statistics

For each rater pair: observed agreement P_o; Cohen's $\kappa = (P_o - P_e)/(1 - P_e)$; positive percent agreement PPA $= 2n_{11}/(2n_{11} + n_{10} + n_{01})$ (the Dice coefficient, insensitive to shared negatives); negative percent agreement NPA $= 2n_{00}/(2n_{00} + n_{10} + n_{01})$. For three-way agreement, Fleiss'

$$\kappa_F = \frac{\bar{P} - \bar{P}_e}{1 - \bar{P}_e}, \qquad \bar{P} = \frac{1}{N}\sum_{i=1}^{N} \frac{\sum_j n_{ij}(n_{ij} - 1)}{k(k-1)}, \qquad \bar{P}_e = \sum_j p_j^2,$$

with N items, $k = 3$ raters, n_{ij} raters assigning category j to item i, and p_j the marginal share of category j. Landis–Koch conventions: 0.61–0.80 substantial, above 0.80 almost perfect. Because graphs are sparse, P_o and NPA are mechanically high; PPA is the demanding measure at the edge level. For low-prevalence codes, κ can be near zero at near-perfect raw agreement (the prevalence paradox); PPA/NPA and prevalence-adjusted κ must be read alongside.

6 Benchmark results

6.1 Cause narratives (live configuration)

Table 3: Pairwise agreement, cause narratives. Model = `gpt-6.1-terra`, live prompt.

		English ($N = 30$)				French ($N = 30$)			
Level	Pair	P_o	κ	PPA	NPA	P_o	κ	PPA	NPA
Edge	Model vs. Coder 1	0.889	0.597	0.662	0.934	0.901	0.593	0.650	0.942
	Model vs. Coder 2	0.921	0.723	0.771	0.952	0.909	0.649	0.703	0.946
	Coder 1 vs. Coder 2	0.895	0.627	0.689	0.937	0.907	0.598	0.652	0.946
Node	Model vs. Coder 1	0.959	0.764	0.786	0.977	0.939	0.715	0.750	0.965
	Model vs. Coder 2	0.972	0.856	0.871	0.984	0.961	0.829	0.851	0.978
	Coder 1 vs. Coder 2	0.957	0.761	0.784	0.976	0.935	0.686	0.723	0.963

Table 4: Three-way Fleiss' κ_F, cause narratives (model, Coder 1, Coder 2)

	English	French
Edge level ($N_{\text{items}} = 514; 772$)	0.659	0.622
Node level ($N_{\text{items}} = 750; 750$)	0.795	0.746

Averaged over the two coders, model–human κ exceeds human–human κ on every row in both languages. At the edge level the model agrees more with Coder 2 than the two coders agree with each other, and slightly less with Coder 1 (by 0.03 in English and less than 0.01 in French). Node-level agreement exceeds edge-level agreement throughout: raters agree on which factors are present and disagree mainly on structure. Inspection of the most disputed edges shows two structural sources: whether to route a factor through #6 (e.g., $8 \to 6$ vs. $8 \to 999$), and substitution among #7, #9, and #21 for general references to industry, capitalism, or production. Rules (b), (d), (j), and (k) address both. One taxonomy gap emerged (population growth), folded into #21.

6.2 Model comparison

We ran the same benchmark with the live prompts on three further OpenAI reasoning models (Table 5). Agreement is stable across models: `gpt-5.4` and `gpt-5.4-mini` are within 0.03 of the live model on every measure, a difference of the size one expects from 30 items per language. The smallest model, `gpt-5-mini`, names the same causes but agrees less on the links between them (0.07–0.09 lower): coding causal structure, not factors, is what demands model capacity. Because each run is tied to its `coding_version` hash, a model change is recorded rather than silent.

Table 5: Three-way Fleiss' κ_F (model, Coder 1, Coder 2) across models, live prompts

	English		French	
	Cause links	Causes	Cause links	Causes
`gpt-6.1-terra` (live)	0.66	0.80	0.62	0.75
`gpt-5.4`	0.64	0.78	0.64	0.75
`gpt-5.4-mini`	0.64	0.79	0.62	0.73
`gpt-5-mini`	0.57	0.77	0.56	0.73

`gpt-6.1-terra` at its default reasoning effort; the other models at medium effort. Cause links: edge level; causes: node level.

7 Full-sample deployment

The coder is applied to every response in the survey; the incremental registry codes only newly added respondents as data collection continues in further countries.

Empty graphs. An empty graph is a valid output. It is returned when a response names no taxonomy factor as a cause, typically because the respondent describes symptoms or evidence of climate change (melting ice, extreme weather) or cites only generic pollution, which factor #6 explicitly excludes. Such responses are a substantive category—respondents who conflate effects with causes—rather than coding noise, and are retained in the analysis.

Causal-chain structure. Every root-to-outcome path in each respondent's DAG is enumerated (root = a node with zero in-degree in that respondent's graph; the five natural-cause factors and the three skepticism factors are merged into one node each). Path length is not mutually exclusive across respondents: a narrative may contain both a one-step and a two-step chain. Comparing how often a factor appears on any edge with how often it is the root of a chain separates factors that respondents name as ultimate causes from factors they use as mechanisms (e.g., greenhouse gases, through which respondents route other causes). This distinction is only observable because the instrument returns structure rather than mentions.

8 Reproducibility and repository

The accompanying repository contains:

- the pipeline, the web interface, and the human-annotation application;
- the live prompt and the taxonomy as plain text;
- the agreement script and the exact commands used to compute every table in this appendix;
- synthetic example data (responses, two human codings and a model output) on which the coder and the agreement script run end to end.

Survey responses, human codings and model outputs are not released. Researchers apply the workflow to their own corpus and benchmark: the coder, the annotation application and the agreement script are the same programs that produced the tables here. Model outputs are non-deterministic across calls, so re-running the model on comparable data reproduces agreement up to sampling variation. An open-weights replication of the benchmark, proposed in the main text, will be added to the same repository under its own `coding_version`.